Statistical Inference and Multivariate Analysis (MA324)

LECTURE SLIDES

Topic 03

Multiple Linear Regression

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Multiple Linear Regression

- In general, the response (y) may be related to p regressors (input variables/predictors).
- The model

$$y_i = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip} + \epsilon_i, i = 1, \ldots, n$$

  is called **multiple linear regression**. A regression model with p regressors.
- The parameters $\beta_j, j = 0, 1, 2, \cdots, p$ are called regression coefficients.
- $\beta_j, j = 0, 1, 2, \cdots, p$ represents the change in the average value of the response for a unit change in j^{th} regressor keeping other regressors fixed.

- Data (of sample size n): $(y_i, x_{i1}, x_{i2}, \cdots, x_{ip}), i = 1, \ldots, n$
- Scalar notation becomes cumbersome – so matrix notation is used
- Let

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \; X = \begin{pmatrix} 1 & x_{11} & \ldots & x_{1p} \\ \vdots & \vdots & x_{ij} & \vdots \\ 1 & x_{n1} & \ldots & x_{np} \end{pmatrix}, \; \beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}, \; \epsilon = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}$$

- Then we can write the model in a more compact form:

$$y_{n\times 1} = X_{n\times(p+1)}\beta_{(p+1)\times 1} + \epsilon_{n\times 1}$$

- X is called the ***design matrix***

Error Assumptions

- As before, ϵ_i's satisfy
  - $E(\epsilon_i) = 0$ for $i = 1, \ldots, n$
  - $Var(\epsilon_i) = \sigma^2$ for $i = 1, \ldots, n$
  - $Cov(\epsilon_i, \epsilon_j) = 0$ for all $i \neq j$
- Equivalently, the assumptions can be written as
  - $E(\boldsymbol{\epsilon}) = \mathbf{0}$
  - $Var(\boldsymbol{\epsilon}) = \sigma^2 I_n$

Estimation of Regression Coefficients:

- Consider the objective function

$$
\begin{aligned}
Q(\boldsymbol{\beta}) &= Q(\beta_0, \beta_1, \cdots, \beta_p) \\
&= \sum_{i=1}^{n} \Big(y_i - \beta_0 - \beta_1 x_{i1} - \ldots - \beta_p x_{ip} \Big)^2 \\
&= (y - X\beta)^T (y - X\beta) \\
&= \big(y^T y - 2\beta^T X^T y + \beta^T X^T X \beta \big)
\end{aligned}
$$

- The LSEs of $\beta_0, \beta_1, \cdots, \beta_p$ are $\underset{\boldsymbol{\beta}}{\operatorname{argmin}}\, Q(\boldsymbol{\beta})$.

Vector Derivative : Linear Function

- Let $\boldsymbol{x}$ be an $n \times 1$ column vector (the variable).
- Let $\boldsymbol{a}$ be a constant $n \times 1$ column vector.
- The derivative of a scalar-valued linear function with respect to the vector $\boldsymbol{x}$ is

$$\frac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{a}^T \boldsymbol{x}) = \boldsymbol{a}.$$

Vector Derivative: Quadratic Function

- Let A be a constant $n \times n$ matrix.
- The derivative of a quadratic form with respect to the vector x is

$$\frac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{x}^T A \boldsymbol{x}) = \left(A + A^T\right) \boldsymbol{x}.$$

- If A is a symmetric matrix, then the above formula simplifies to

$$\frac{\partial}{\partial \boldsymbol{x}}\left(\boldsymbol{x}^T A \boldsymbol{x}\right) = 2A\boldsymbol{x}.$$

Estimation of Model Parameters:

- Differentiating $Q(\beta)$ with respect to β and setting the derivative to zero gives the following *normal equations*:

$$X^T X\beta = X^T y.$$

- The above system of linear equations in $\boldsymbol{\beta}$ is always consistent.

- If the matrix $X^T X$ is *invertible* (i.e. if X is of *full rank*), then the LSE is unique and given by

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

Fitted Value:

- The fitted value of response corresponding to regression $x = (1, x_1, \cdots, x_p)$ is

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \ldots + \hat{\beta}_p x_p$$

- Then $\hat{y} = (\hat{y}_1, \hat{y}_2, \cdots, \hat{y}_n)^T = X\hat{\beta} = X(X^T X)^{-1} X^T y = Hy$, where, $H = X(X^T X)^{-1} X^T$ is called **hat-matrix.**

- The residuals:

$$\boldsymbol{e} = \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} y_1 - \hat{y}_1 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix} = \boldsymbol{y} - \hat{\boldsymbol{y}} = (I - H)\boldsymbol{y}$$

Properties of LSE of β

- $\hat{\beta}$ is a linear function of y
- $\hat{\beta}$ is unbiased estimator of β. That is, $E(\hat{\beta}) = \beta$
- $Var\left(\hat{\beta}\right) = \sigma^2(X^T X)^{-1}$
- $\hat{\beta}$ is the BLUE of β.